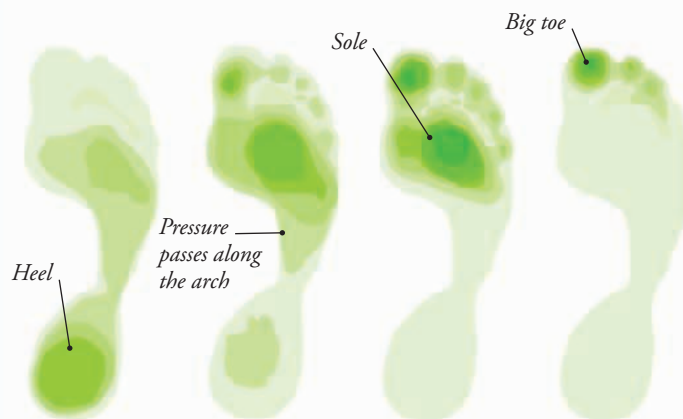




ONE STEP AT A TIME

With each step you take, you push down first through your heel, then the springy sole, and finally your big toe, which lifts your body off the ground. This sequence of footprints shows the weight shift from heel to toe. The darker the green color, the more body weight is pushing downward.



A group of four powerful muscles—the quadriceps—straighten your knee when you kick.

These strong tendons keep the kneecap in place and allow your leg to flex and twist.

Your kneecap protects your knee joint.

WOW!

The average person will walk about 80,000 miles (128,000 km) in a lifetime. That's the same as walking around the world three times.

ON TIPTOE

When you stand on tiptoe, the calf muscles at the back of your lower leg pull the heel bone upward so that your foot is supported only by the toes. Dancing *en pointe* (on tiptoe) like a ballet dancer exerts a lot of pressure on your toes because your whole body weight presses down on a very small area of your body.



► **EN POINTE** In this ballet position, the tendons on top of the foot are stretched to a maximum.

IN ACTION

Your legs are the most muscular parts of your body. If you exercise or play sports, these muscles can become incredibly powerful. A professional soccer player, for example, can kick a ball with enough force to make it travel at about 70 mph (110 kph).

► KICKING A SOCCER BALL

This sequence shows what happens when a ball is kicked. Your hamstring muscles (red) pull the leg back. Then the thigh is pulled forward by the quadriceps (green), so that the lower leg can flick forward to kick the ball.

